Data Mining and Machine Learning

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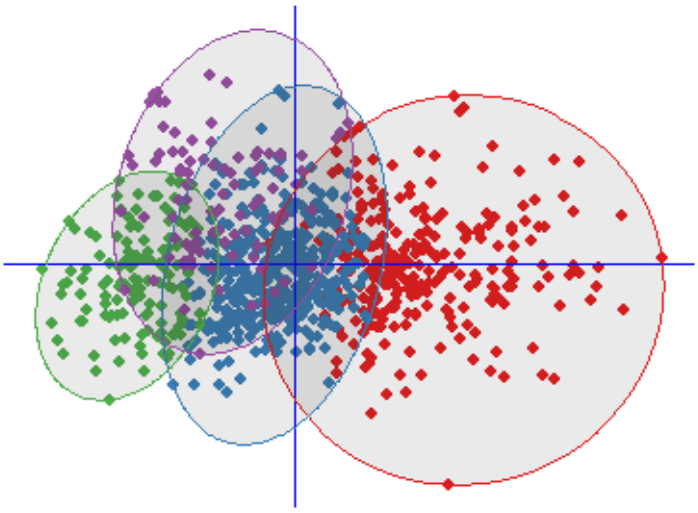

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1 Probability Model: The Foundation of Inference

1.1 Descriptive vs. Inferential Statistics

In Data Science, we deal with two fundamentally different problems. It is crucial to distinguish them immediately.

Descriptive Statistics (The Direct Problem)

We observe the **entire population** (e.g., every single customer in a database).

- **Goal:** Summarize the data exactly.
- **Result:** Certain facts (e.g., *The exact average age is 35.2*).
- **Uncertainty:** Zero.

Statistical Inference (The Indirect Problem)

We only observe a small **sample** (e.g., 100 surveyed customers).

- **Goal:** Infer the characteristics of the whole population based on the sample.
- **Result:** Estimates with uncertainty (e.g., *The avg age is likely between 34 - 36*).
- **Tool:** We use Probability to measure that uncertainty.

Basic Definitions. To move from a sample to a population, we must formally define the entities involved and the goals of inference.

- **Population**: the entire group under study to which conclusions are to be applied.
- **Sample**: a subset of the population that is actually observed and collected.
- **Statistical Inference**: The process of drawing conclusions about the population from data collected on a sample, using probability reasoning.

1.1.1 The Two Main Tasks of Inference

As a Data Scientist, you will mostly perform two tasks:

1. **Estimation:** This involves estimating unknown population characteristics (parameters) based on sample statistics.
   - **Point Estimation**: Assigning a single value to the population parameter (e.g., using the sample mean $\bar{x}$ to estimate the population mean μ).
   - **Interval Estimation**: Assigning an interval of plausible values (a confidence interval) to the parameter.
2. **Hypothesis Testing:** Deciding whether a claim (hypothesis) about a population parameter is acceptable or should be rejected, in probabilistic terms.(e.g., Did the new marketing campaign actually increase sales or was it just luck?).

Example 1.1. Parameter Estimation.

A Data Scientist wants to know the average age (μ) of all customers of an e-commerce platform (Population). Since they cannot survey millions of users, they take a sample of $n = 100$ customers.

- `Step1 (Descriptive)`:
  - Calculate the sample mean age. Suppose the sample mean is $\bar{x} = 35.2$ years.
- `Step2 (Point Estimation)`:
  - The point estimate for the population mean μ is 35.2.
- `Step3 (Interval Estimation)`:
  - Based on sample var and size, they calculate a 95% Confidence Interval for μ, say $[34.0, 36.4]$. They infer that the true μ age of *all* customers is likely between 34.0 and 36.4 y.o.

1.1.2 The Importance of Unbiased Random Samples

Since observations are made only on a sample, the fundamental questions are:

- How to extend sample conclusions to the population?
- What are the limits of validity and reliability of these conclusions?

The solution lies in using an **Unbiased Random Sample**:

- A sample where the units are selected through a **random draw**, ensuring that the selection process is not influenced by the studied variable. This means every unit in the population has the same probability of being chosen.
  - This method is the best approach to **ensure representativeness and validity of inference**.
  - If the survey is not random, inferential methods cannot be applied because the sample might not accurately reflect the population.

1.2 Probability: The Measure of Uncertainty

1.2.1 Why Do We Need Probability?

In the previous section, we learned that we often only have a **Sample**, but we want to know about the **Population**. Because the sample is just a small piece of the whole, our conclusions are never 100% certain.

Probability is the mathematical tool we use to measure that uncertainty. It allows us to say: *I am 95% confident that the average age is between 34 and 36,* rather than just guessing.

1.2.2 The Three Definitions of Probability

How do we actually define *Chance*? There are three main schools of thought. As a Data Scientist, you will mostly use the **Classical** and **Frequentist** approaches, but **Subjectivist** is the basis of Bayesian Statistics.

1. **Classical Probability (Gambling):** Used when all outcomes are equally likely (e.g., dice, cards).
$$P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$$
2. **Frequentist Probability (Simulation):** Used for repeatable experiments. Probability is the **relative frequency** of an event happening over a huge number of trials.
$$P(E) \approx \frac{\text{Count of Successes}}{\text{Count of Trials}} \quad (\text{as trials } \to \infty)$$
3. **Subjectivist Probability (Belief):** Used when experiments cannot be repeated (e.g., *Will humanity go to Mars by 2030?*). It measures a degree of personal belief or betting odds.

Example 1.2. Example: Tossing Coins

If you toss two coins, the Sample Space is $\{HH, HT, TH, TT\}$. The event `Getting exactly one head` has 2 favorable outcomes (HT, TH).

$$P(\text{One Head}) = \frac{2}{4} = 0.5$$

1.3 Axioms and Fundamental Rules

1.3.1 Set Theory and Venn Diagrams

We treat Probability mathematically using **Set Theory**. We visualize outcomes using **Venn Diagrams**.

- **Trial**: is a single execution of an experiment or process that results in an outcome (e.g.,The single roll of the die).
- **Sample Space (S):** the set of all possible outcomes of an experiment. The rectangle containing everything that can happen.
- **Event (A):** a collection of outcomes, or a subset of the sample space S. A circle inside the rectangle representing a specific outcome. Events are combined using Boolean Logic Operators: Union $\cup$, Intersection $\cap$, Complement $\overline{A}$.

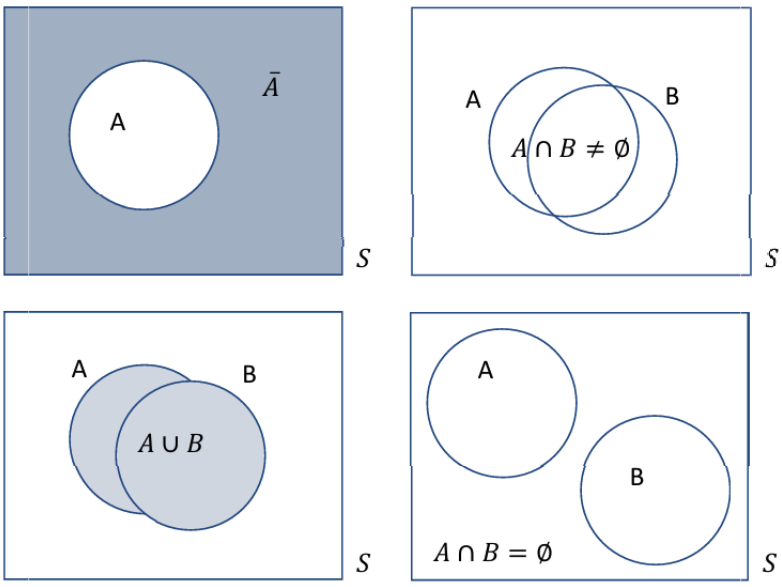

Figure 1.3.1: *Venn Diagram*

1.3.2 Kolmogorov's Three Axioms

The **three fundamental axioms** define the mathematical properties that any probability measure P must satisfy.

1. **Positivity:** You cannot have a negative chance.

$$P(E) \geq 0$$

2. **Certainty:** The probability that *something* in the sample space happens is 100%.

$$P(S) = 1$$

3. **Additivity (for Disjoint Events):** If two events cannot happen at the same time (Mutually Exclusive, $A \cap B = \emptyset$), you just add their probabilities.

$$P(A \cup B) = P(A) + P(B)$$

1.3.3 Important Theorems derived from Axioms

From those three simple rules, we derive the formulas you will use in practice.

- **The Complement Rule**: The probability that an event E does *not* occur ($\overline{E}$) is one minus the probability that E occurs. This implies that $0 \leq P(E) \leq 1$.

$$P(\overline{E}) = 1 - P(E) \tag{1}$$

- **The General Addition Law (Union):** If events A and B **can** overlap (happen together), we cannot just add them, or we would double-count the overlap. We must subtract the intersection.

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

- **Probability of the Empty Set ($\emptyset$)**: The probability of the impossible event is zero. This is derived from Axiom 3 (Additivity) and Axiom 2 (Unit Measure).

$$P(\emptyset) = 0 \tag{2}$$

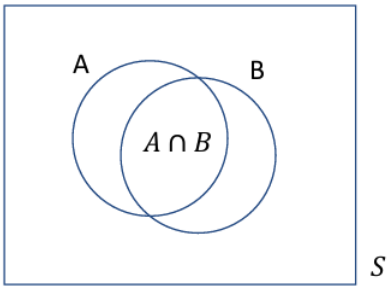

Figure 1.3.2: *General Addition Law*

Example 1.3. Example: Passing Exams

Let $P(Stat) = 0.60$ and $P(CS) = 0.50$. If 35% of students pass **both** ($P(Stat \cap CS) = 0.35$), what is the chance of passing **at least one**?

$$P(Stat \cup CS) = 0.60 + 0.50 - 0.35 = 0.75$$

1.3.4 Extension: Three Events

If you have three overlapping events (A, B, C), the logic is the same: Add the single circles, subtract the pairwise overlaps, and add back the center (because you subtracted it too many times).

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

1.4 Conditional Probability

This is the most critical concept for understanding how Machine Learning models (like Naive Bayes) work. It answers: *How do my chances change if I get new information?*

1.4.1 Defining Conditional Probability

Conditional Probability

The probability of event B happening, **given that** we already know event A has happened.

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \tag{3}$$

We shrink the sample space. We no longer care about the whole world (S), only the world where A is true.

The Factorization Theorem (Total Probability)

If we want to find the total probability of B, we sum up all the ways B can happen (via A or via $\bar{A}$):

$$P(B) = P(A \cap B) + P(\bar{A} \cap B) = P(B|A)P(A) + P(B|\bar{A})P(\bar{A}) \tag{4}$$

1.4.2 Stochastic Independence

Two events are **Independent** if knowing A gives us zero information about B.

$$P(B|A) = P(B) \tag{5}$$

If (and only if) this holds, we can just multiply probabilities: $P(A \cap B) = P(A) \cdot P(B)$.

Example 1.4. Example 1: The Two-Way Table

Let's apply this to a real dataset. A = Female, B = Wears Glasses.

Use Glasses? / Gender	$\bar{B}$	B	*Total*
$\bar{A}$	$P(\bar{A} \cap \bar{B}) = 0.2$	$P(\bar{A} \cap B) = 0.4$	$P(\bar{A}) = 0.6$
A	$P(A \cap \bar{B}) = 0.2$	$P(A \cap B) = 0.20$	$P(A) = 0.4$
Total	$P(\bar{B}) = 0.4$	$P(B) = 0.6$	$P(S) = 1$

Figure 1.4.1: *Probability Two way table*

- **Question:** What is the probability a person wears glasses (B), *given* they are Female (A)?
- **Calculation:** Look only at the *Female* row.

$$P(B|A) = \frac{20 \text{ (Females with glasses)}}{40 \text{ (Total Females)}} = 0.5$$

- **Check Independence:** $P(B) = 60/100 = 0.6$. Since $P(B|A) \neq P(B)$ ($0.5 \neq 0.6$), Gender and Glasses are **Dependent**.

1.4.3 Bayes' Theorem

This is the most important formula in Data Science. It tells us how to *update our beliefs* when we see new evidence.

Bayes' Theorem

$$P(A|E) = \frac{P(E|A) \cdot P(A)}{P(E)} \tag{6}$$

- $P(A)$: **Prior**. What we believed *before* seeing data.
- $P(E|A)$: **Likelihood**. How probable is the evidence if our hypothesis is true?
- $P(A|E)$: **Posterior**. Our new belief *after* seeing the evidence.

1.4.4 Real World Application: Medical Diagnosis

This is a classic interview question to test if you understand *Base Rate Fallacy*.

Example 1.5. The False Positive Paradox Scenario:

- A rare disease affects 1% of people ($P(D) = 0.01$).

- The test is 95% accurate ($P(+|D) = 0.95$).
- The test has a 10% false positive rate ($P(+|\text{Health}) = 0.10$).

Question: You test positive. What is the probability you actually have the disease?

The Calculation (Bayes):

$$P(D|+) = \frac{P(+|D) \cdot P(D)}{P(+)}$$

1. **Numerator (True Positive path):** $0.95 \times 0.01 = 0.0095$
2. **Denominator (Total Positives):** True Positives + False Positives

$$(0.95 \times 0.01) + (0.10 \times 0.99) = 0.0095 + 0.099 = 0.1085$$

3. **Result:**

$$\frac{0.0095}{0.1085} \approx \mathbf{8.76\%}$$

Conclusion: Even with a positive test, you are likely healthy! Why? Because the disease is so rare (1%) that the false alarms (10% of the 99% healthy people) drown out the real cases.

1.5 Combinatorial Calculus and Discrete Random Variables

1.5.1 Combinatorial Calculus

Before calculating probabilities, we often need to count how many ways something can happen. This is **Combinatorics**.

Ordered Sequences (Does Order Matter? YES)

1. With Replacement: You have n options, and you choose k times. After each choice, you put the item back.

$$n^k \tag{7}$$

- *Example:* A 4-digit PIN code ($10^4 = 10,000$ codes).

2. Without Replacement (Permutations): You have n options, choose k. Once picked, an item is gone

$$D_{n,k} = \frac{n!}{(n-k)!} \tag{8}$$

- *Example:* A race with 8 runners. Who gets Gold, Silver, and Bronze ($k = 3$)?

Unordered Sequences (Does Order Matter? NO)

Combinations (The Binomial Coefficient): You choose k items from n, and the order does **not** matter.

$$C_{n,k} = \binom{n}{k} = \frac{n!}{k!(n-k)!} \tag{9}$$

- *Example:* Choosing a team of 3 Data Scientists from a department of 8.

This is the most critical formula for Data Science (used in sampling).

1.5.2 Discrete Random Variables

A **Random Variable** (X) is just a function that translates a *real world outcome* into a number. It is **Discrete** because can only take specific integer values (e.g., number of sales, number of clicks).

Example 1.6. Discrete Random Variable

Imagine an experiment, like flipping a coin three times. The raw outcomes (elementary events) are: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.

The Discrete Random Variable (X) is defined as the "Number of Heads" in the three flips: 3, 2, 2, ...

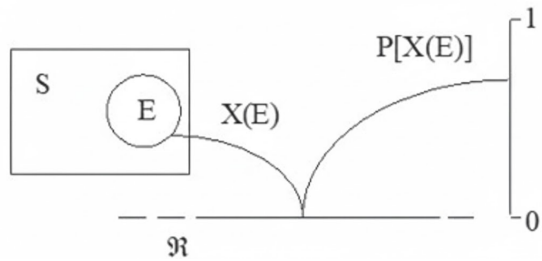

Figure 1.5.1: *Discrete Random variable*

Key Metrics: Mean and Variance

1. Expected Value (mean): If you repeated the experiment infinitely, what would be the average?

$$E(X) = \mu = \sum x_i \cdot p_i \tag{10}$$

2. Variance: How far are the values typically from the mean?

$$\text{Var}(X) = E[(X - \mu)^2] \tag{11}$$

1.5.3 Discrete Probability Models

In the previous section, we learned what a Random Variable is (a way to map outcomes to numbers). Now we'll explore several important **probability distributions** that model common real-world scenarios.

Think of these as *patterns* that describe how data behaves in the real world. Once you recognize the pattern (e.g., *counting successes*), you can just pick the right tool (e.g., *Binomial*) and use its ready-made formulas.

1. The Basic Building Blocks

Uniform Distribution $U(k)$

Scenario: *Everything is equally likely.*

- **Example:** Rolling a fair die.
- **Probability:** $P(X = x) = \frac{1}{k}$
- **Mean:** The exact middle value.

Bernoulli Distribution $B(1, \pi)$

Scenario: *The Atom of Probability.* A single trial with only two outcomes: Success (1) or Failure (0).

- **Example:** A user clicks an ad (Yes/No).
- **Parameter π:** The probability of success.
- **Mean:** π.
- **Variance:** $\pi(1 - \pi)$ (Variance is highest when $\pi = 0.5$: maximum uncertainty).

2. Counting Successes (The *Binomial* Family)

These distributions answer the question: *How many successes did I get?*

Binomial Distribution $B(n, \pi)$

Scenario: You repeat a Bernoulli trial n times independently.

- **Example:** 100 users visit your site ($n = 100$), and each has a 5% chance of buying ($\pi = 0.05$). How many sales (X) do I get?
- **Formula:** Uses the combination $\binom{n}{x}$.
- **Mean:** $n\pi$ (Expected sales $= 100 \times 0.05 = 5$).

Hypergeometric Distribution $H(N, n, m)$

Scenario: Same as Binomial, but **Sampling Without Replacement**.

- **Example:** You have a deck of 52 cards (N), with 4 Aces (m). You draw 5 cards (n). How many are Aces?
- **Key Difference:** The trials are **dependent**. If you draw an Ace first, it's harder to draw another one.
- **Usage:** Important for Quality Control (sampling from small batches).

3. Waiting Time (The *Geometric* Family)

These distributions answer the question: *How long do I have to wait?*

Geometric Distribution $G(\pi)$

Scenario: How many trials until the **first** success?

- **Example:** How many job interviews until I get an offer? (assuming 20% success rate).
- **Mean:** $1/\pi$ (If success rate is 20%, expect to wait 5 interviews).
- **Property:** *Memoryless.* If you failed 10 times, your probability on the 11th try is exactly the same.

Negative Binomial $NB(k, \pi)$

Scenario: How many trials until the k**-th** success?

- **Example:** How many cold calls to make 5 sales?
- It is essentially the sum of k independent Geometric variables.

4. Rate of Events (Poisson)

Poisson Distribution $P(\lambda)$

Scenario: Modeling rare events over a fixed time or space.

- **Example:** Number of server crashes per month. Number of customers arriving per hour.
- **Parameter** λ**:** The average rate of occurrence.
- **Unique Property:** Mean = Variance = λ.
- **Usage:** Crucial for queuing theory and reliability engineering.

Random Variable	Probability Distribution	Expectation	Variance
Uniform $X \sim U_d(k)$	$P(X = x) = 1/k \quad for\ x = 1, \dots, k$	$E(X) = \frac{k+1}{2}$	$Var(X) = \frac{(k^2-1)}{12}$
Bernoulli $X \sim B(1, \pi)$	$P(X = x) = \pi^x (1-\pi)^{1-x} \quad for\ x = 0,1$	$E(X) = \pi$	$Var(X) = \pi(1-\pi)$
Binomial $X \sim B(n, \pi)$	$P(X = x) = \binom{n}{x} \pi^x (1-\pi)^{1-x}$ $for\ x = 0,1, \dots, n$	$E(X) = n\pi$	$Var(X) = n\pi(1-\pi)$
Pascal or Negative Binomial $X \sim NB(k, \pi)$	$P(X = x) = \binom{x-1}{k-1} \pi^k (1-\pi)^{x-k}$ $for\ x = k, k+1, \dots$	$E(X) = k\frac{1}{\pi}$	$Var(X) = k\frac{1-\pi}{\pi^2}$
Poisson $X \sim P(\lambda)$	$P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}$ $for\ x = 0,1, \dots$	$E(X) = \lambda$	$Var(X) = \lambda$
Hypergeometric $X \sim H(N, n, m)$	$P(X = x) = \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}}$ $for \max(0, n-N+m) \le x \le \min(n, m)$	$E(X) = n\frac{m}{N}$	$Var(X) = n\frac{m}{N}\frac{N-n}{N-1}$

Figure 1.5.2: *Discrete probabilty model tabel*